# Breast Cancer Diagnosis

Predicting malignancy from cell nucleus measurements with supervised and unsupervised learning

MU5BM753 Supervised and Unsupervised learning for biology

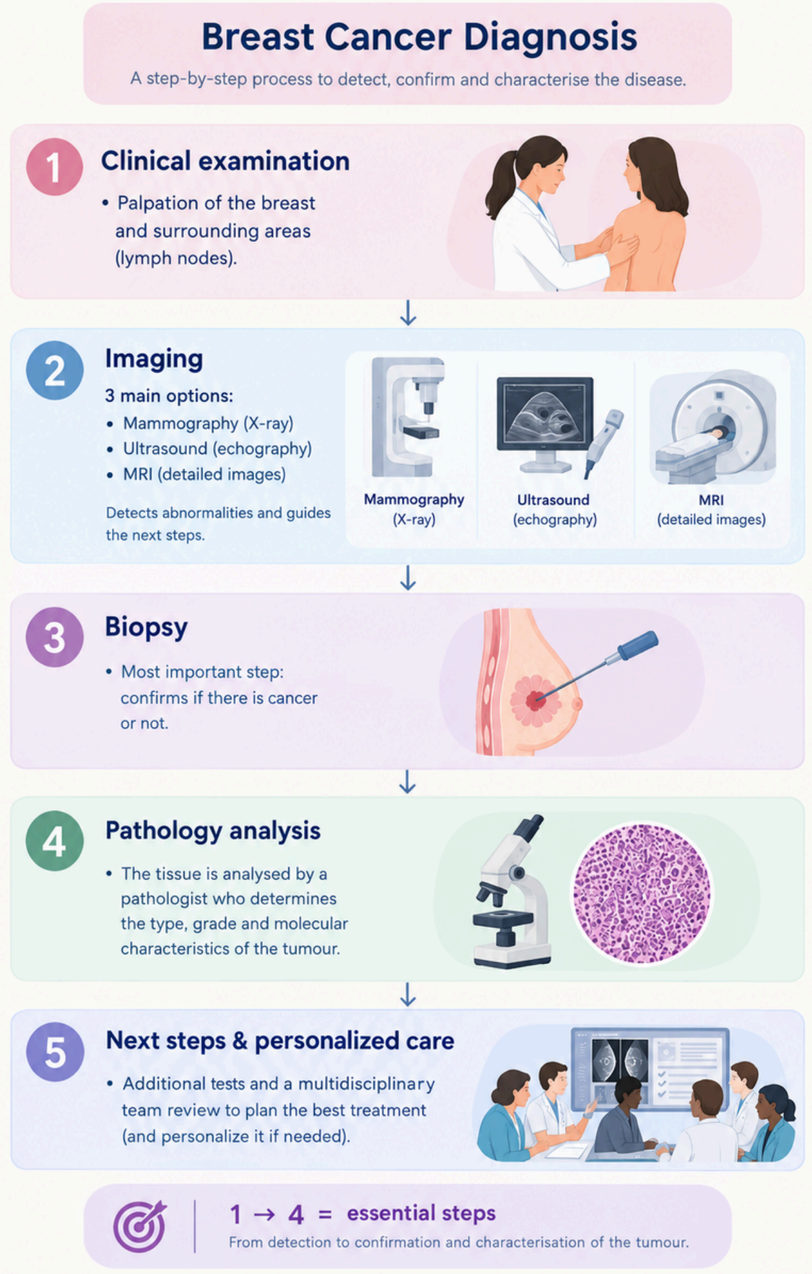
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# Breast cancer : overview

- **Mammary epithelial cells** acquire alterations that **disrupt normal** control of proliferation, survival, genome maintenance and tissue organization
- These changes also modify the **cell's phenotype**, in particular, **nuclear appearance**
- Malignant cells typically show:
  - Enlarged, irregular nuclei
  - Greater variation in nuclear size and shape
  - Disrupted chromatin patterns

➔ **cytological criteria used by pathologist**[1,2]

- **Diagnosis**: made by a pathologist reading the slide by eye
- **Limitation**: human assessment can vary between pathologists[3] and takes a long time
- **Goal**: use nuclear features for a fast, quantitative assessment



# One hypothesis, tested two ways

➔ **Nuclear morphology alone carries enough signal to reveal and predict malignancy**

## Supervised approach

Can a classifier reliably :

- **Help the pathologist** by providing a **rapid preliminary assessment** of the likelihood of malignancy ?
- **Support the diagnostic** assessment ?
- **identify** which nuclear characteristics drive that prediction the most ?

Can quantified nuclear atypia (size, contour, texture, ...) offer the pathologist a more reproducible second opinion ?

## Unsupervised approach

Are the **nuclear characteristics** alone sufficient to separate most of the samples into **distinct groups**, particularly benign and malignant tumors ?

Is malignant transformation a strong enough signal on its own to create two distinct, separable cell populations ?

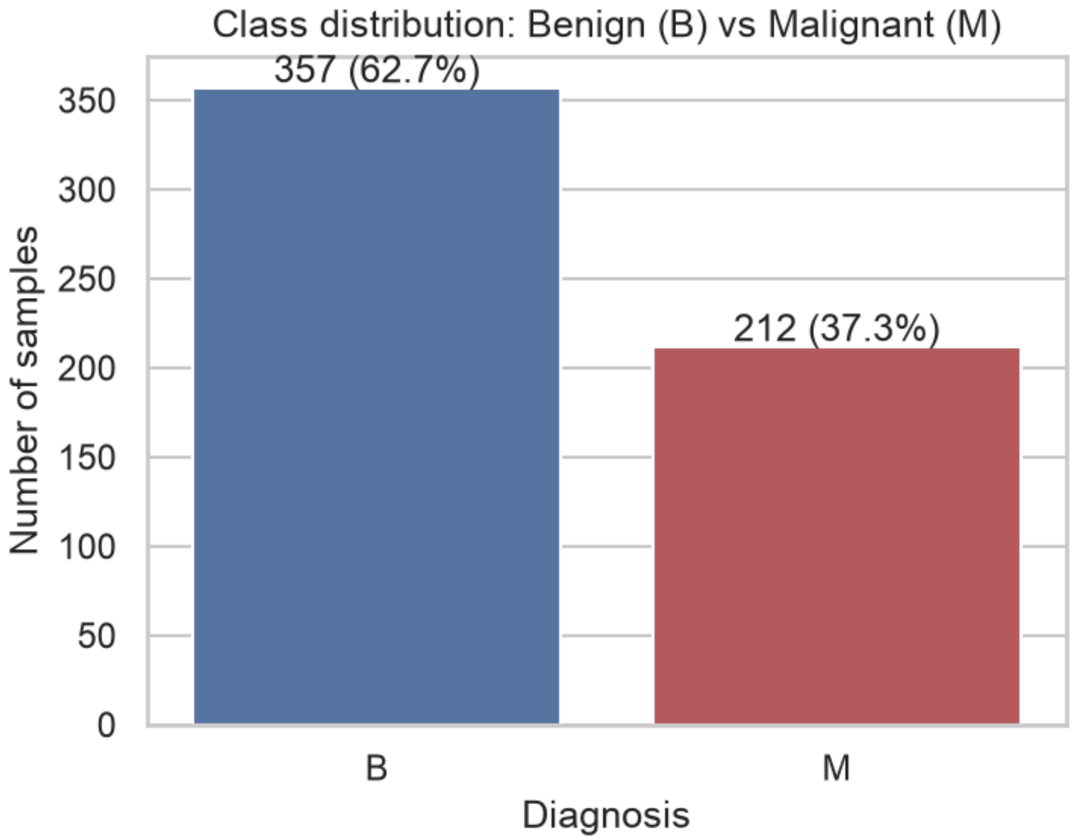
# The Wisconsin Breast Cancer dataset

**569**
tumor samples

**30**
measured features

- Each sample: **10 base measurements** of a cell nucleus (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension),
- Each is summarized by 3 statistics: mean, standard error, and worst (the 3 most extreme cells).
- The classes are imbalanced: 63% benign, 37% malignant.

Quality checks confirm the data is clean: no missing values, no duplicate samples.

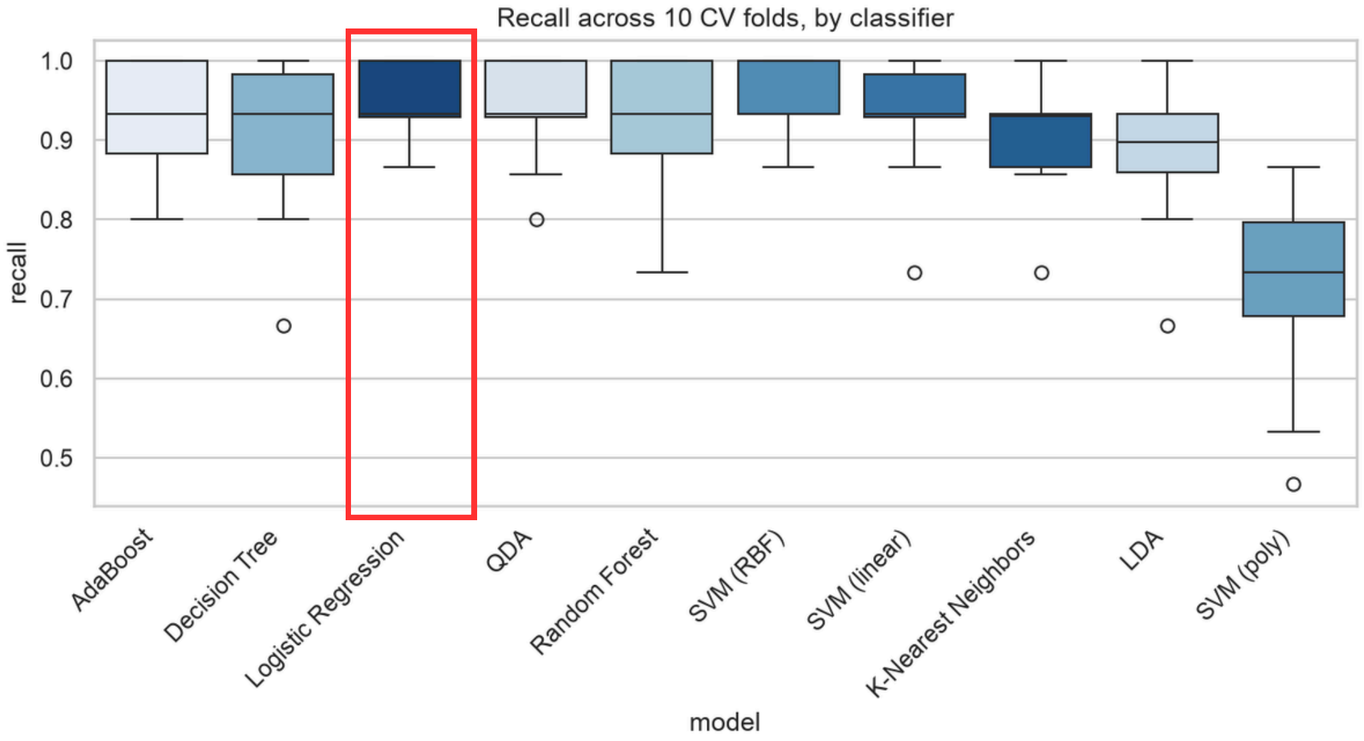


# Bioinformatic workflow

1. **Quality checks**
2. **Exploratory Analysis**
3. **Initial Screening**
elects models by **average rank** and confirmed by **recall/ARI**
4. **Feature selection**
Different methods used
5. **Supervised approach**
6. **Unsupervised approach**

**Average rank :** ranks each model per metric, then averages
**Recall :** proportion of malignant cases caught
**ARI :** agreement with the true diagnosis, corrected for chance

# Logistic Regression is the strongest supervised baseline




| | model | accuracy_mean | precision_mean | recall_mean | f1_mean | f2_mean | roc_auc_mean | avg_rank_across_6_metrics |
|---|---|---|---|---|---|---|---|---|
| 0 | Logistic Regression | 0.977308 | 0.986607 | 0.952381 | 0.968440 | 0.958569 | 0.992610 | 2.00 |
| 1 | SVM (RBF) | 0.974872 | 0.980000 | 0.952857 | 0.965475 | 0.957691 | 0.992667 | 2.67 |
| 2 | AdaBoost | 0.967372 | 0.980833 | 0.932381 | 0.954261 | 0.940643 | 0.994305 | 3.67 |

- **10 classifiers** are compared with 10-fold **cross-validation** on 6 metrics

**~0.95**

**Mean Recall** for the **Logistic Regression**

**2.00**

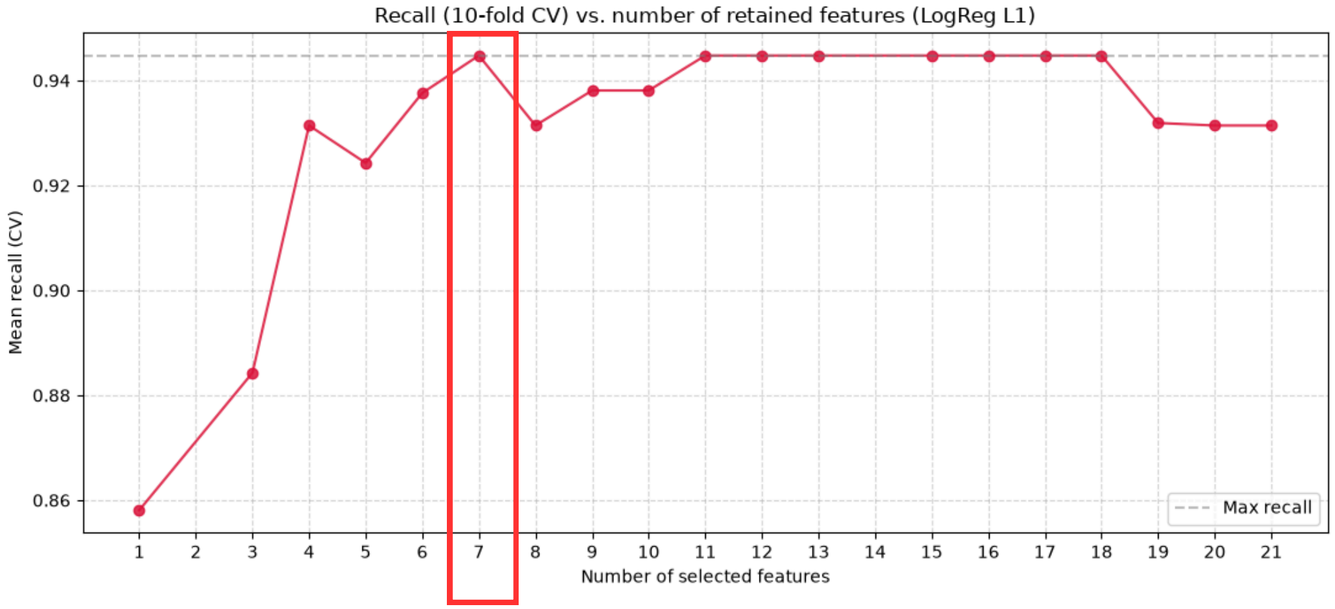
**Average rank** across 6 metrics for **Logistic Regression**

This **screening** does **two things** at once:

- **Picks** which classifier to deploy
- **Fixes** which model will drive embedded feature selection next

# 7 of 30 features are enough to predict malignancy

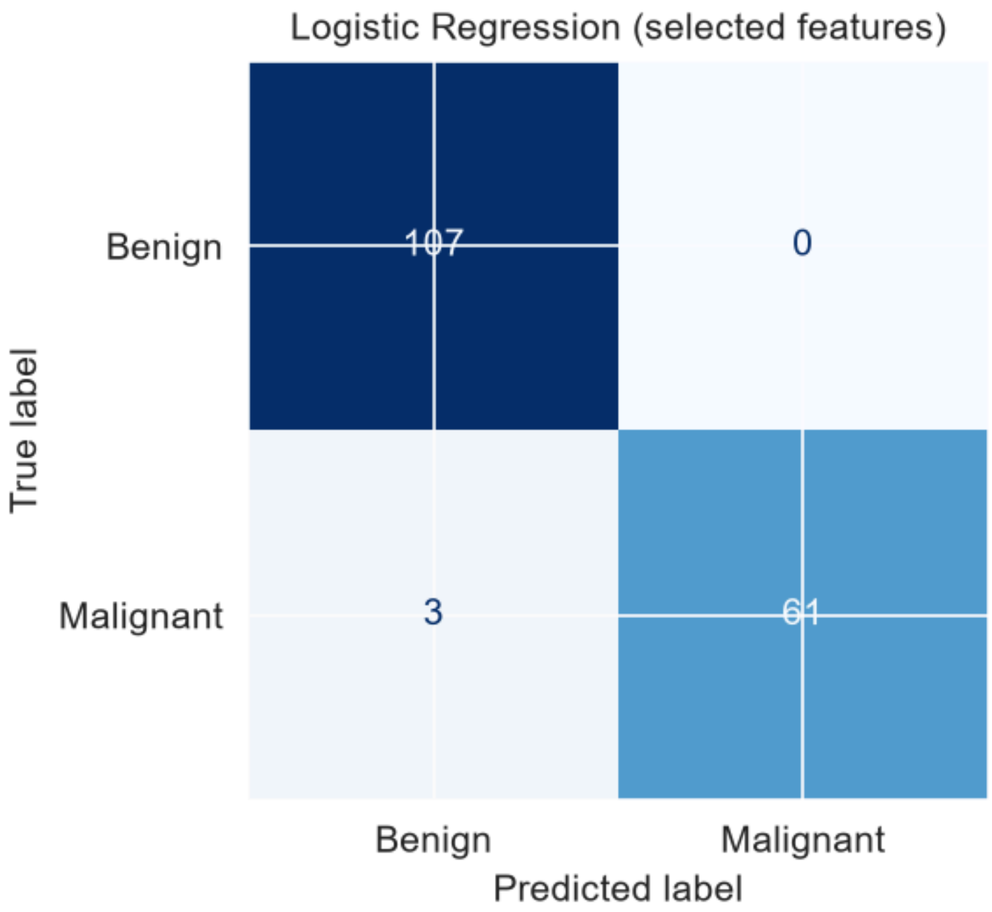
| Method | Family | Features | Recall | Avg. rank |
|---|---|---|---|---|
| LogReg L1 | embedded | 7 | 94.5% | 2.0 |
| SelectFdr | filter | 18 | 93.8% | 2.4 |
| LinearSVC L1 | embedded | 9 | 93.8% | 2.9 |
| Elastic-Net | embedded | 6 | 93.8% | 3.8 |
| Variance Threshold | filter | 23 | 93.1% | 4.0 |




- **7 collinear features** (correlation above 0.95) are **removed** first.
- 5 **selection methods** are then benchmarked, from two families:
  - **Filter** : scores each feature on its own.
  - **Embedded** : selects features while training a model.
- **LogReg L1** wins : best **average rank** (2.0) and strongest **recall** (94.5%).

**The 7 selected features :**
radius_mean, concave_points_mean, radius_se, texture_worst, smoothness_worst, concave_points_worst, symmetry_worst

# Tuned model misses only 3 malignant cases out of 64




- Tuned by grid search using the 1-SE rule (C = 0.3857) on the 7 selected features.
- Evaluated on the held-out test set, never seen during training or tuning.

**95.3%**

**Recall:** only 3 malignant cases missed out of 64

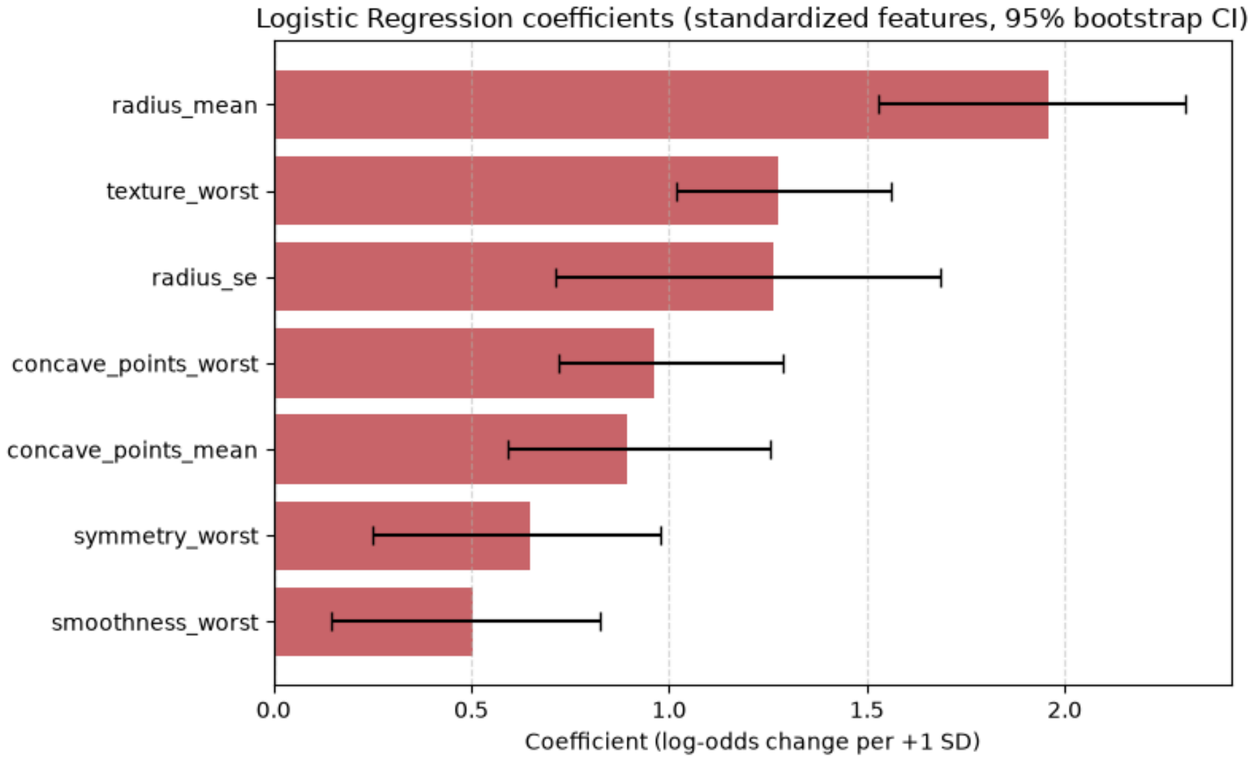
**100%**

**Precision:** no false alarm among 61 flagged cases

A **classifier** reliably predicts malignancy from a **selected** set of **measurements**.

# Nuclear Size and Texture Drive the Model's Decision

| | coefficient | ci_low | ci_high | odds_ratio_per_SD |
|---|---|---|---|---|
| **radius_mean** | 1.960 | 1.528 | 2.306 | 7.096 |
| **texture_worst** | 1.274 | 1.018 | 1.563 | 3.576 |
| **radius_se** | 1.263 | 0.711 | 1.687 | 3.537 |
| **concave_points_worst** | 0.964 | 0.720 | 1.288 | 2.622 |
| **concave_points_mean** | 0.895 | 0.593 | 1.257 | 2.448 |
| **symmetry_worst** | 0.649 | 0.248 | 0.977 | 1.914 |
| **smoothness_worst** | 0.504 | 0.145 | 0.827 | 1.656 |

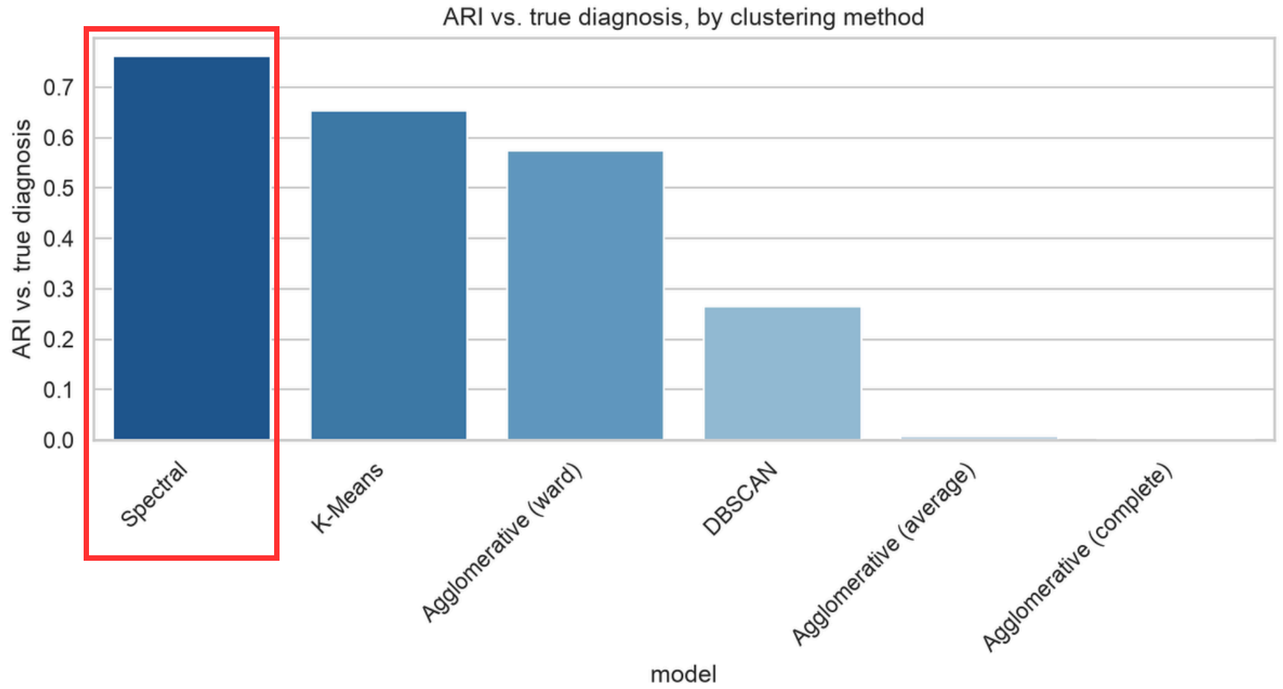



- Log-odds coefficients evaluated on standardized data.
- 95% Confidence Intervals validated via 1,000 bootstrap iterations.

**Dominant driver :** *radius_mean*
Increases the odds of malignancy by a factor of 7.1 per additional standard deviation.

**Reliable features**
No confidence interval crosses zero. Every selected feature actively contributes to the final prediction.

# Spectral Clustering recovers best the true diagnosis




| model | silhouette | homogeneity | completeness | v_measure | ARI_vs_diagnosis | avg_rank_across_5_metrics |
|---|---|---|---|---|---|---|
| Spectral | 0.336653 | 0.649723 | 0.676197 | 0.662696 | 0.760801 | 1.8 |
| K-Means | 0.343382 | 0.524888 | 0.540147 | 0.532408 | 0.653625 | 2.2 |
| omerative (ward) | 0.339385 | 0.446214 | 0.468142 | 0.456915 | 0.575041 | 3.2 |

- **6 clustering methods** group the samples without ever using the diagnosis
- Their output is then **checked** against the true labels using **ARI**

**0.76**

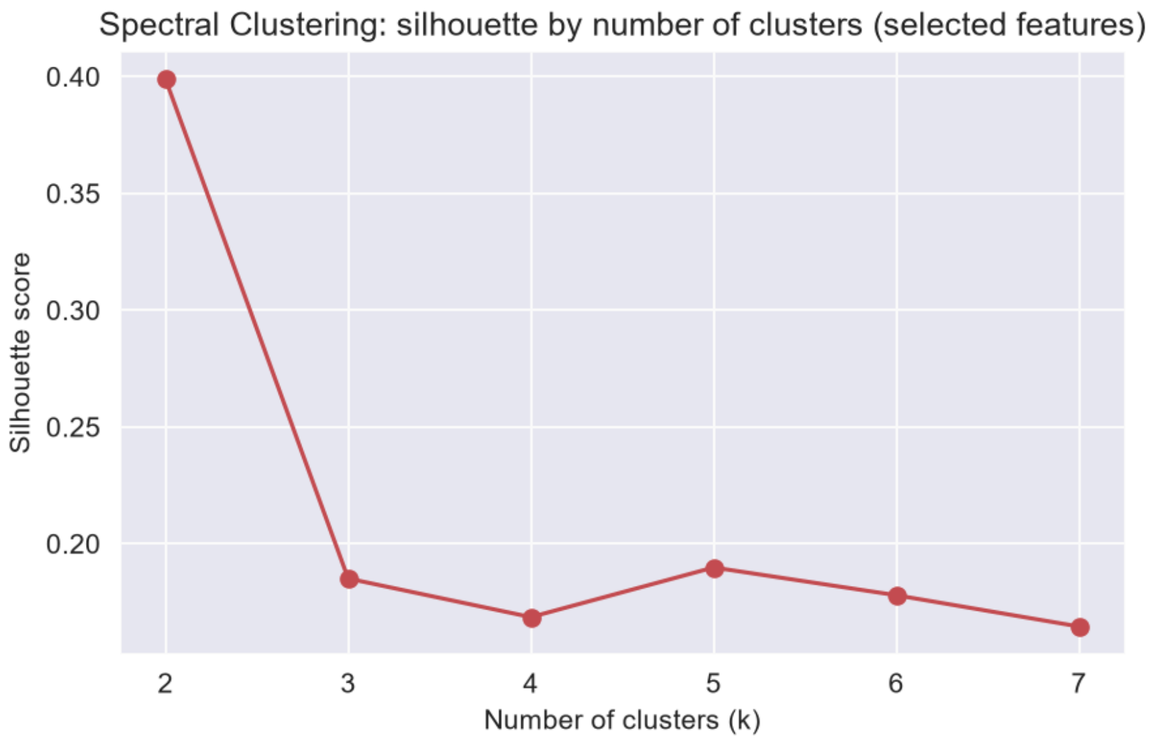
**Highest ARI** for **Spectral clustering**

**1.8**

**Average rank** across 5 metrics for **Spectral clustering**

- This **screening picks** which clustering method to deploy.

# Two clusters capture the main structure



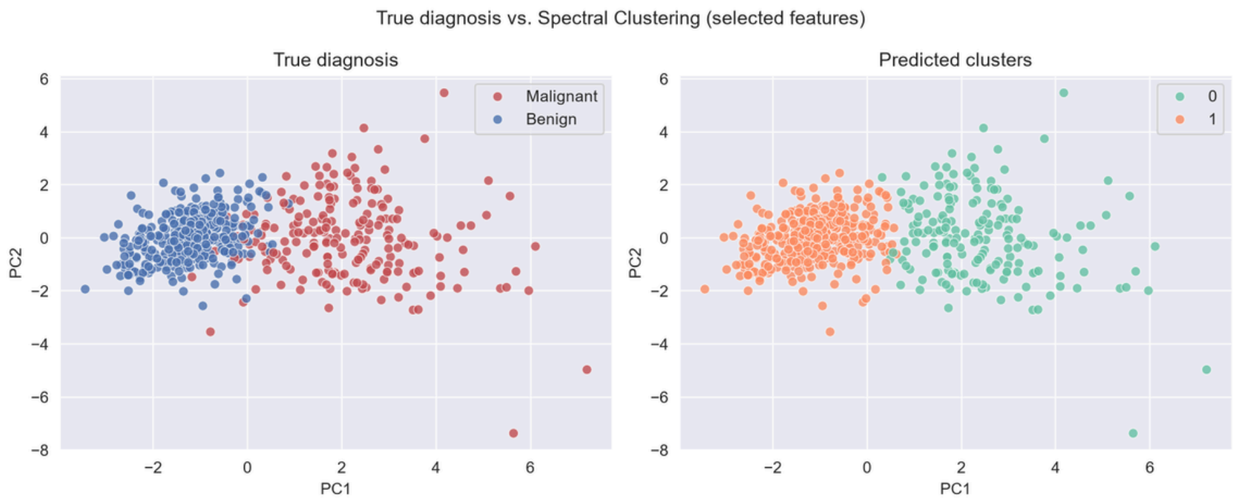

- The **screening** above **assumed** 2 groups, matching the 2 known diagnosis classes.
- Here, k is swept from 2 to 7 using **silhouette**

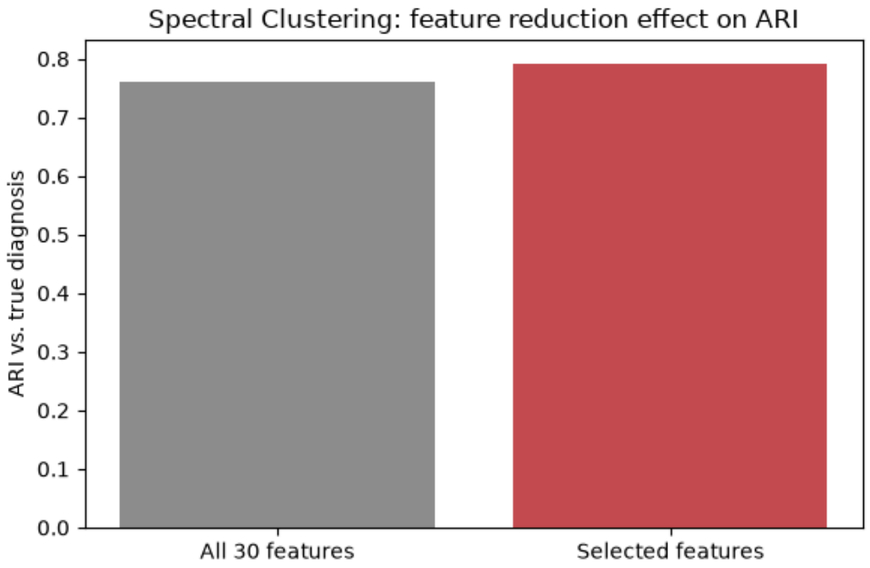
## Silhouette peaks sharply at k = 2

0.4 at k = 2, dropping to 0.16–0.19 for every k from 3 to 7. More clusters reveal no additional structure.

- The **data** itself **confirms** the **assumption**, without ever seeing the labels:
  - **malignant** and **benign nuclei** each form **one coherent group**,
  - with **no sign of hidden sub-types** within either class.

# Feature selection strengthens the clustering–diagnosis match




- The **PCA projection** shows the predicted clusters closely matching the true diagnosis.
- Restricting to the **7 selected features** raises the ARI from **0.761** to **0.792**

**0.792**

Adjusted Rand Index versus the true diagnosis, without the clustering ever seeing the labels

**Malignancy** alone is a **strong enough signal** to organize the data into 2 groups, without ever using the labels.

# The 7 features mirror criteria pathologists already use

**Size**
(**radius_mean, radius_se**)

- Malignant nuclei tend to be larger and more heterogeneous in size[6]
- Cancer cells can have abnormal chromosome numbers (aneuploidy) which can increase DNA content

**Contour irregularity**
(**concave_points_mean, concave_points_worst, smoothness_worst**)

- These features capture how irregular the nuclear boundary is
- Irregular nuclear outlines and pleomorphism are important criteria for assessing malignancy

**Symmetry & texture**
(**symmetry_worst, texture_worst**)

- Abnormalities in the shape and distribution of pixel intensities within the nucleus
- Differences in chromatin organization and nuclear appearance

4 of the 7 features are "worst" statistics: malignancy shows up most clearly in a sample's most abnormal-looking cells, not its average shape.

# Strong results still come with limitations

**Same Blind Spot as Pathologists**

- Some invasive cancers keep regular, benign-looking nuclei — invisible to pure geometry, whether by model or by eye.
- The samples both methods missed (false negatives, misclassified clusters) fit exactly this profile.

➔ a second opinion for clear cases, not the ambiguous ones

**Single-Institution Generalizability**

Features are derived from a localized, specific imaging setup. The model's exact mathematical thresholds require multi-center validation on modern equipment before any broader clinical use.

# External validation is the natural next step

**Testing the 7-feature model on the WPBC cohort**

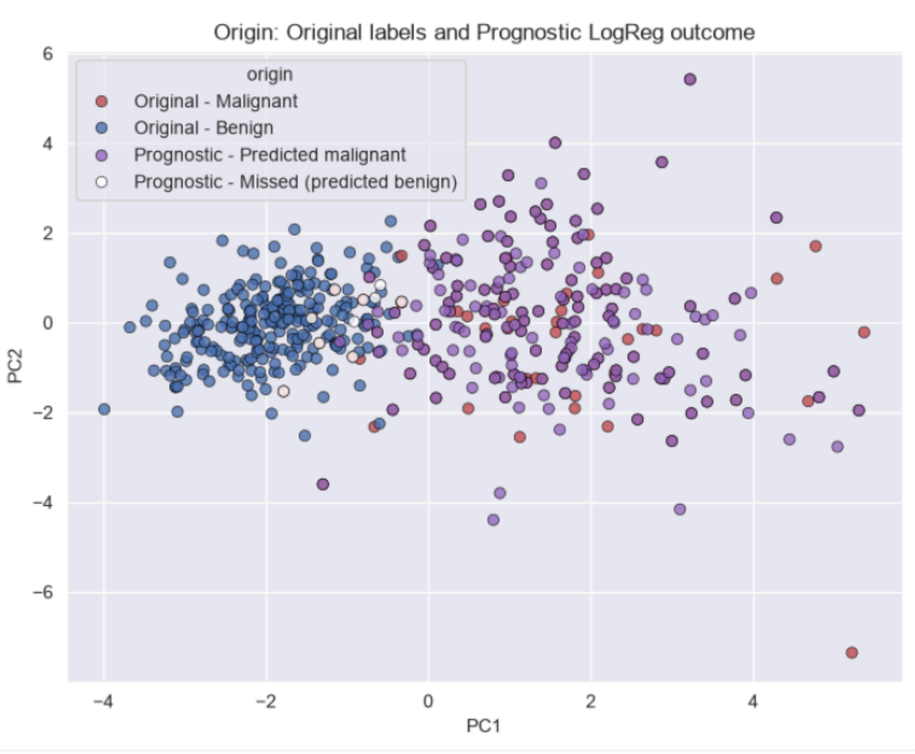
- **Mathematical limitations:** Absence of benign samples precludes the calculation of *precision*, *specificity*, and *false-positive rates*.

- **Analysis of the 13 missed cases:**
  - **Morphological overlap:** False negatives mathematically and visually mimic healthy tissues.
  - **Independent Confirmation:** This benign resemblance is strictly confirmed by both the model's decision weights and unsupervised clustering outputs.

**Next Step, Diagnosis to Prognosis**: Transitioning from binary diagnosis to predictive tracking by correlating morphological deviations with outcome (recurrence) and time-to-recurrence.

**93.4%**

Recall: only 13 malignant cases missed out of 198



# One hypothesis confirmed two ways

## Supervised approach

**Logistic Regression**, tuned on 7 selected features :

- misses only **3 malignant cases out of 64** (recall 95.3%),
- with **no false alarm** (precision 100%).

## Unsupervised approach

**Spectral Clustering** recovers :

- **2 natural groups** that closely match the diagnosis (ARI 0.792), without ever using the labels
- and **testing more clusters** finds no further structure.

- **Nuclear morphology** alone is enough to both **predict** and reveal malignancy
- Reducing it to 7 non-redundant features, out of 30, sharpens both views at once

➔ they align with criteria pathologists already use.

**However :**

- **Both methods** struggle with **borderline cases**, just like **pathologists**
- They can **support diagnosis** in **clear-cut cases**, but do not replace expert judgment in difficult cases

# References

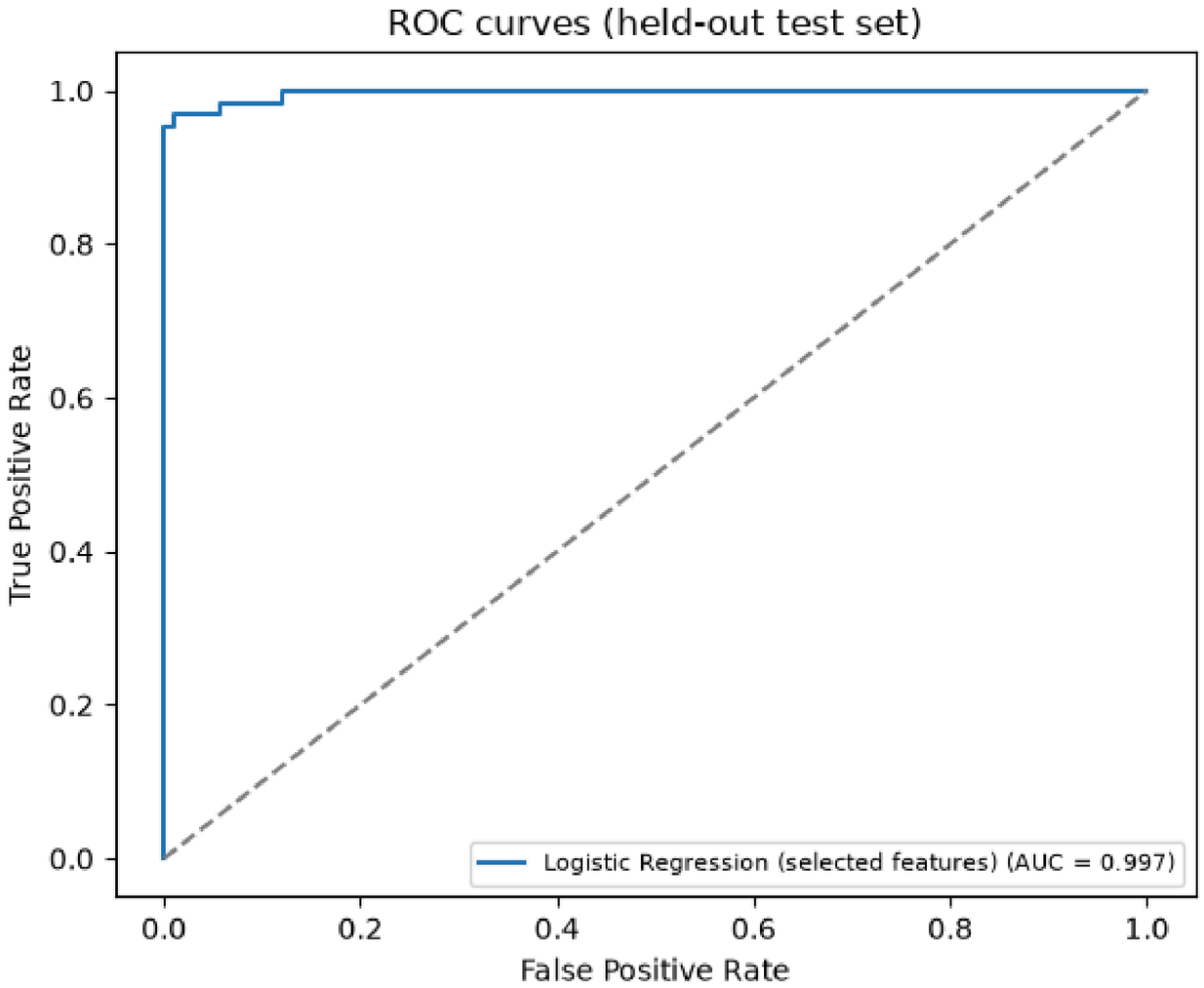
**1.** National Cancer Institute. How Breast Cancer Is Diagnosed. cancer.gov/types/breast/diagnosis
**2.** Bussolati, G., Marchiò, C., Gaetano, L., Lupo, R., & Sapino, A. (2008). Pleomorphism of the nuclear envelope in breast cancer: a new approach to an old problem. Journal of Cellular and Molecular Medicine, 12(1), 209–218.
**3.** Elmore, J. G., Longton, G. M., Carney, P. A., et al. (2015). Diagnostic concordance among pathologists interpreting breast biopsy specimens. JAMA, 313(11), 1122–1132.
**4.** Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. Proceedings of SPIE, 1905, 861–870.
**5.** UCI Machine Learning Repository. Breast Cancer Wisconsin (Diagnostic). doi.org/10.24432/C5DW2B
**6.** Schirmer, E. C., & De Las Heras, J. I. (Eds.). (2014). *Cancer Biology and the Nuclear Envelope: Recent Advances May Elucidate Past Paradoxes*. Advances in Experimental Medicine and Biology, Vol. 773. Springer New York. https://doi.org/10.1007/978-1-4899-8032-8

# Appendix

# Features

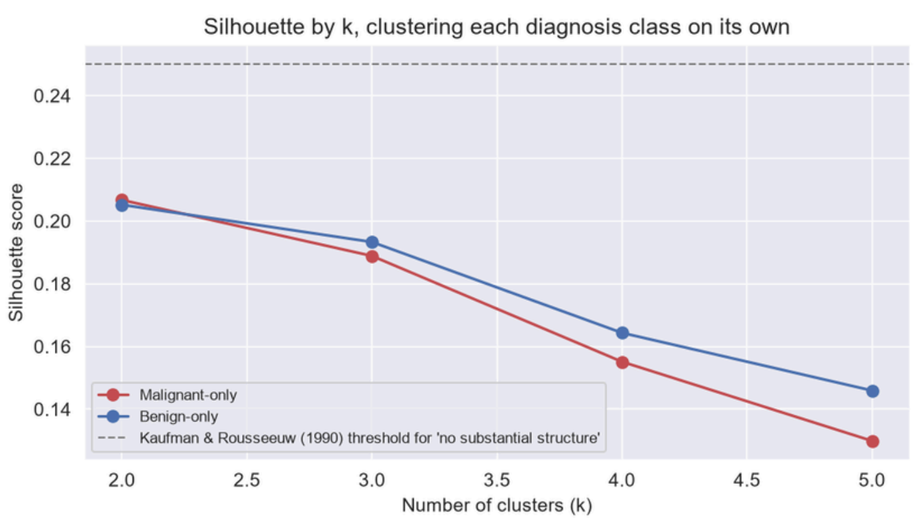
| Feature | What it measures |
|---|---|
| `radius` | Mean distance from the center of the nucleus to its perimeter |
| `texture` | Standard deviation of gray-scale values. Rougher surfaces give higher values |
| `perimeter` | Total length of the nucleus's outer contour |
| `area` | Total 2D surface occupied by the nucleus |
| `smoothness` | Local variation in radius length. Measures edge regularity |
| `compactness` | perimeter^2 / area - 1.0. Measures how compact or round the shape is |
| `concavity` | Severity of the concave, indented portions of the contour |
| `concave points` | Number of concave portions of the contour |
| `symmetry` | How symmetric the nucleus shape is |
| `fractal dimension` | Structural complexity and irregularity of the contour |

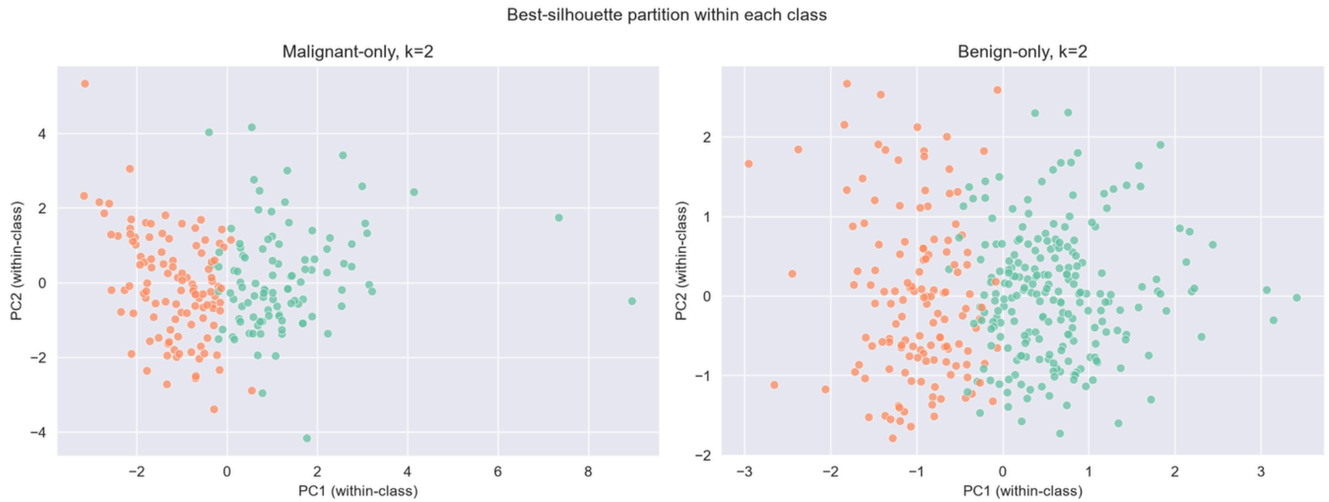
# Tuned Logistic Regression Model




ROC-AUC: **0.997**. Near-perfect ranking of malignant above benign cases, across every threshold.

# Splitting each class alone confirms it: no hidden sub-type



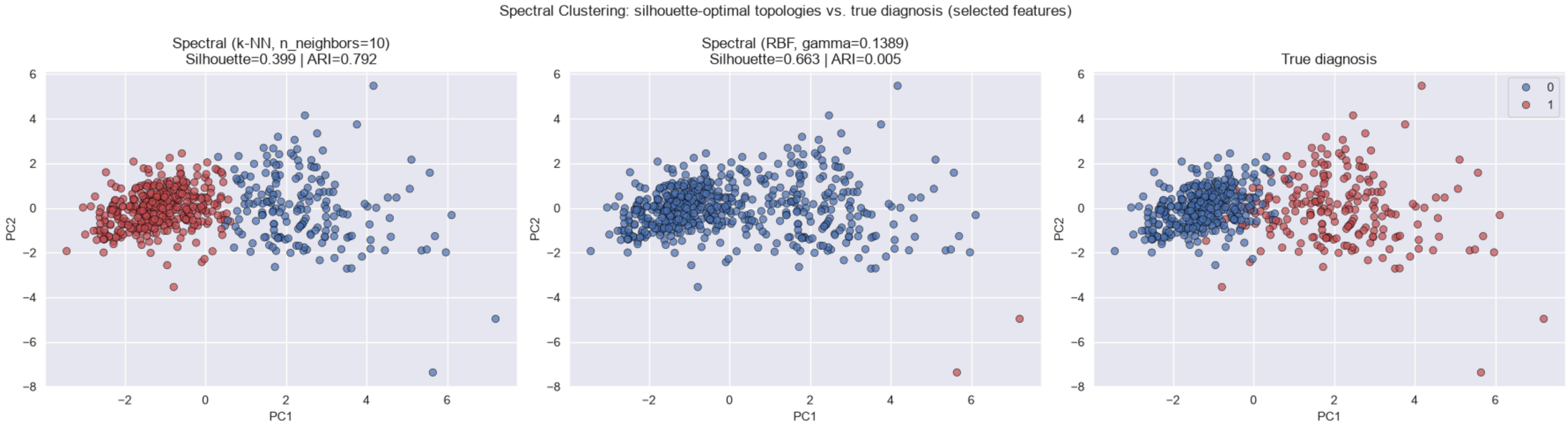




The check above uses the whole dataset. Here, malignant and benign samples are clustered separately, removing the dominant diagnosis signal that could mask a weaker, within-class sub-type.

**Both land on a modest split (k = 2)**
Silhouette 0.207 (malignant) and 0.205 (benign) ➔ both below 0.25, the threshold Kaufman & Rousseeuw (1990) use for "no substantial structure".

The **PCA plots** confirm this visually: both partitions run along one axis with heavy overlap and no visible gap — a cut through a single continuous cloud, not two distinct sub-groups.

# RBF's best silhouette hides a clinically meaningless partition




Both affinity families are swept, then each one's own silhouette-best setting is compared on ARI:

**k-NN (n=10)**

**ARI = 0.792**

(silhouette 0.399)

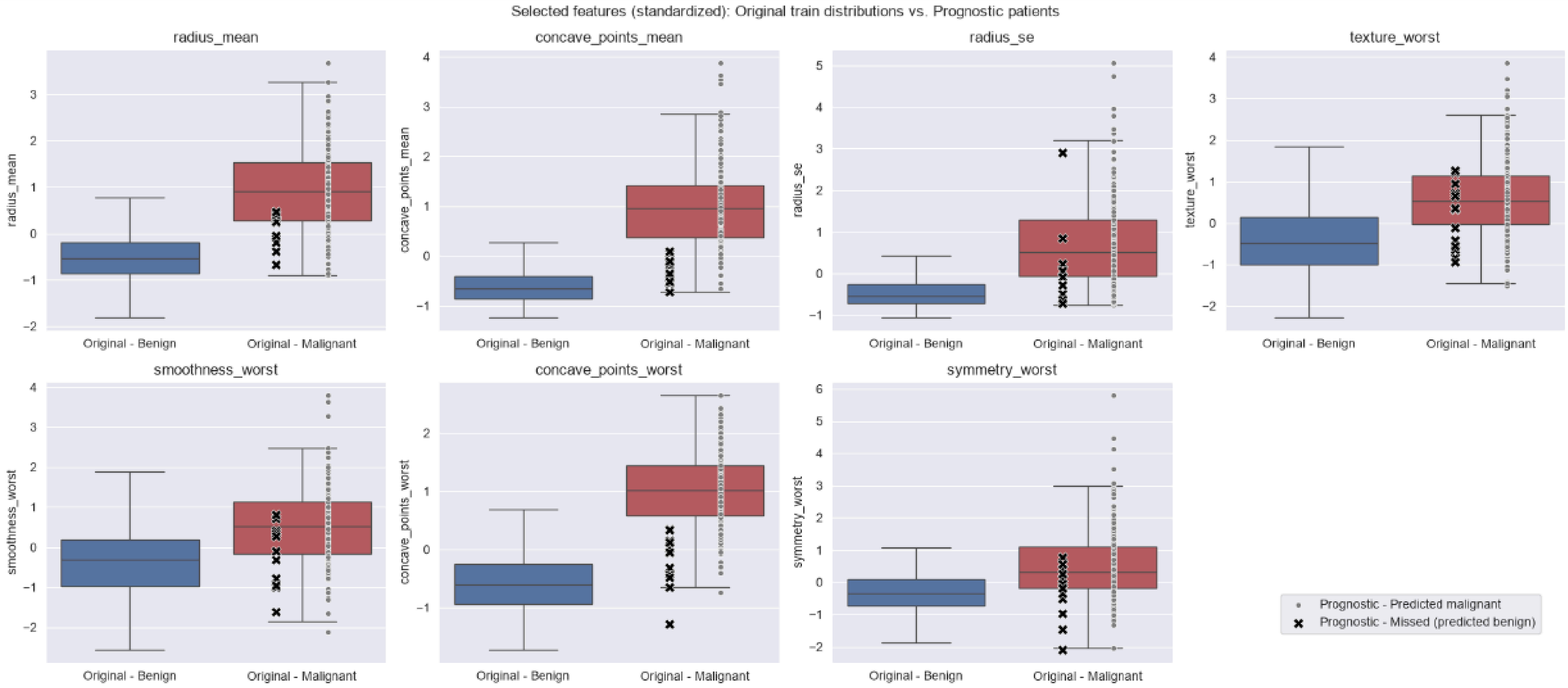
**RBF (γ=0.139)**

**ARI = 0.005**

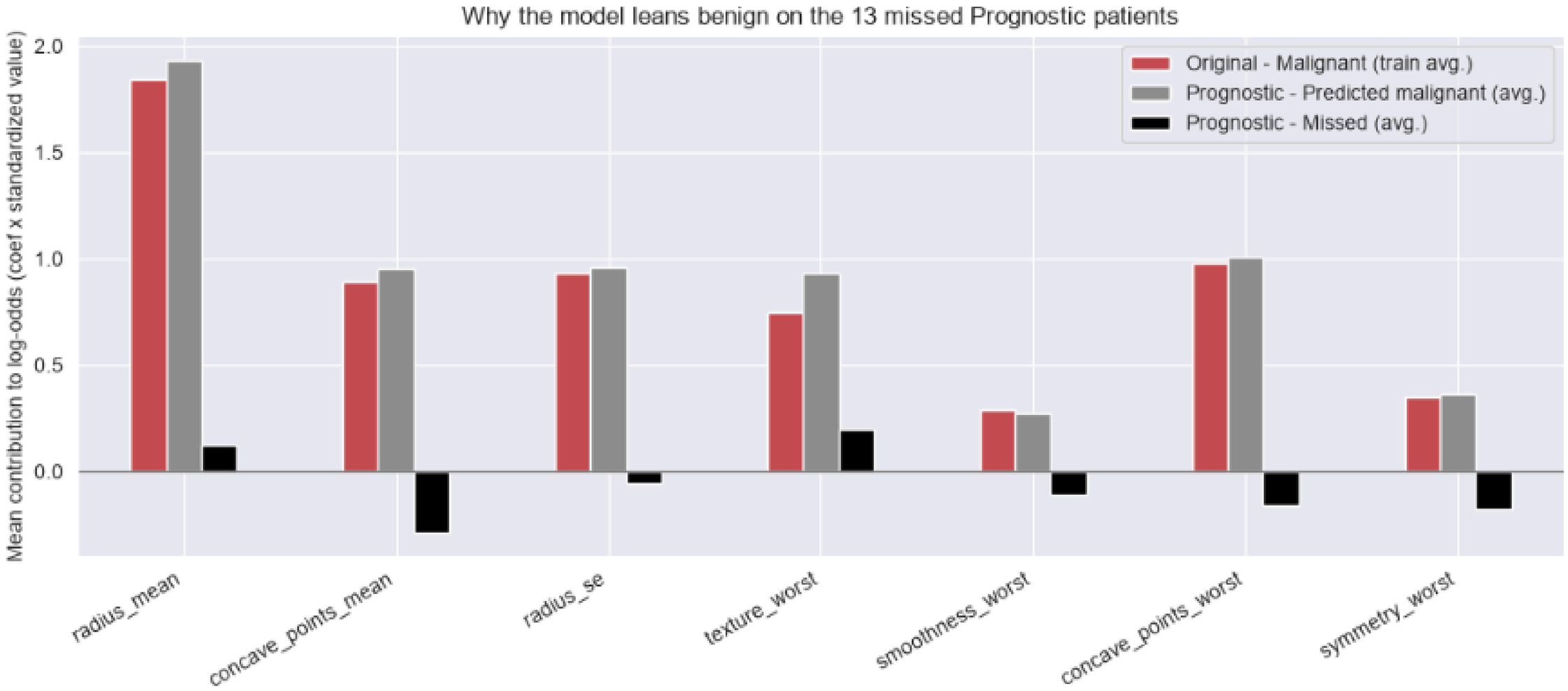
(silhouette 0.663, the highest of the sweep)

RBF's own best silhouette configuration is nearly a random partition against the true diagnosis. k-NN stays stable and clinically aligned across its whole tested range. This is why k-NN, not RBF, is the affinity used throughout.

# Understanding False Negatives in External Validation



# Understanding False Negatives in External Validation (2)



# Implementation details

1. Reusable functions live in `01_scripts/` and are imported by every notebook — no logic is duplicated across the 7 notebooks.

2. The train/test split is made once, in `04_Feature_Selection.ipynb`, and reused unchanged by both "zoom" notebooks — this avoids leaking test information into feature selection or hyperparameter tuning

3. Every random step is fixed (`random_state=42`), so every result in this report is exactly reproducible from the raw data alone.

4. Outputs are separated by type and traceable to the notebook that produced them: tables in `04_results/`, figures in `05_figures/`, filenames prefixed by notebook number.

# Difficulties encountered

**1** **Logistic Regression: the highest-recall model had unstable coefficients :**
The max-recall `C` (30.4) let correlated features split predictive weight unpredictably between them. Adopting the 1-SE rule (C=0.39) traded a negligible drop in cross-validated recall for coefficients that are stable and interpretable.

**2** **External validation needed a scaler that was never saved**
The scaler fit on all 30 features wasn't persisted by `04_Feature_Selection.ipynb`. For the external cohort (WPBC), it had to be reconstructed deterministically — same seed, same split — rather than refit on new data, which would have placed WPBC in a different feature space than the model was trained in.